Oxford College of Emory University

Chapter 13: Fiscal Policy

Part 1: Fiscal Policy and Stabilization

ECON_OX 112 · Principles of Macroeconomics

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The Federal Budget: FY2025

Federal government only · October 2024–September 2025 · Billions of dollars

Revenues	$ billions
Individual income taxes	2,656
Payroll taxes	1,748
Corporate income taxes	452
Customs duties (tariffs)	195
Excise taxes	106
Estate and gift taxes	29
Federal Reserve remittances	5
Miscellaneous fees and fines	42
Total revenues	**5,235**

Outlays	$ billions
Social Security	1,575
Medicare (net)	988
Medicaid	668
Income security programs	397
Other mandatory (net)	539
Defense discretionary	893
Nondefense discretionary	980
Net interest	970
Total outlays	**7,010**

Budget balance: $5,235 − $7,010 = −$1,775 billion (deficit)

A financing shortfall—not a measure of the effect on GDP.

Source: CBO, FY2025 Federal Budget; Revenues; 2026 Outlook. Components may not sum because of rounding. Medicare and other mandatory outlays are net of offsets.

What Does Federal Spending Pay For?

> **Public services**
>
> National defense, federal courts, and law enforcement.

Source: CBO, Federal Budget and spending breakdowns, FY2025

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Retirement and disability benefits, food assistance, and refundable tax credits.

Interest on federal debt

Payments to lenders; these do not buy current goods or services.

Source: CBO, Federal Budget and spending breakdowns, FY2025

Roadmap

Policy goals and instruments

Purchases, household taxes, and transfers

Demand and policy limits

Expansion, contraction, criticisms, and delays

The multiplier

Initial spending and successive consumption rounds

Stabilizers and budget balances

Automatic responses, deficits, and sustainability

Fiscal Policy and Stabilization

Definition

Fiscal policy uses government purchases, taxes, and transfers to influence aggregate demand.

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Stabilization goal

Support output and employment in a downturn; restrain excess demand and inflation pressure.

Expansionary policy shifts AD right; contractionary policy shifts AD left.

Fiscal Instruments

Instrument	Example	Immediate channel
Purchases (G)	Road maintenance	A purchase of current output
Transfers (TR)	Unemployment benefits	Household disposable income
Household taxes (T)	Personal income taxes	Household disposable income

Transfers are not G. Public infrastructure investment is G; I is private investment.

GDP accounting: BEA

Fiscal Policy and Aggregate Expenditure

$$Y = \mathbf{C} + \mathbf{I} + \mathbf{G} + \mathbf{NX}$$

C: Consumption

Household taxes and transfers change disposable income.

Tax cuts or higher transfers can raise C.

I: Private investment

G: Government purchases

NX: Net exports

Fiscal Policy and Aggregate Expenditure

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Fiscal Policy and Aggregate Expenditure

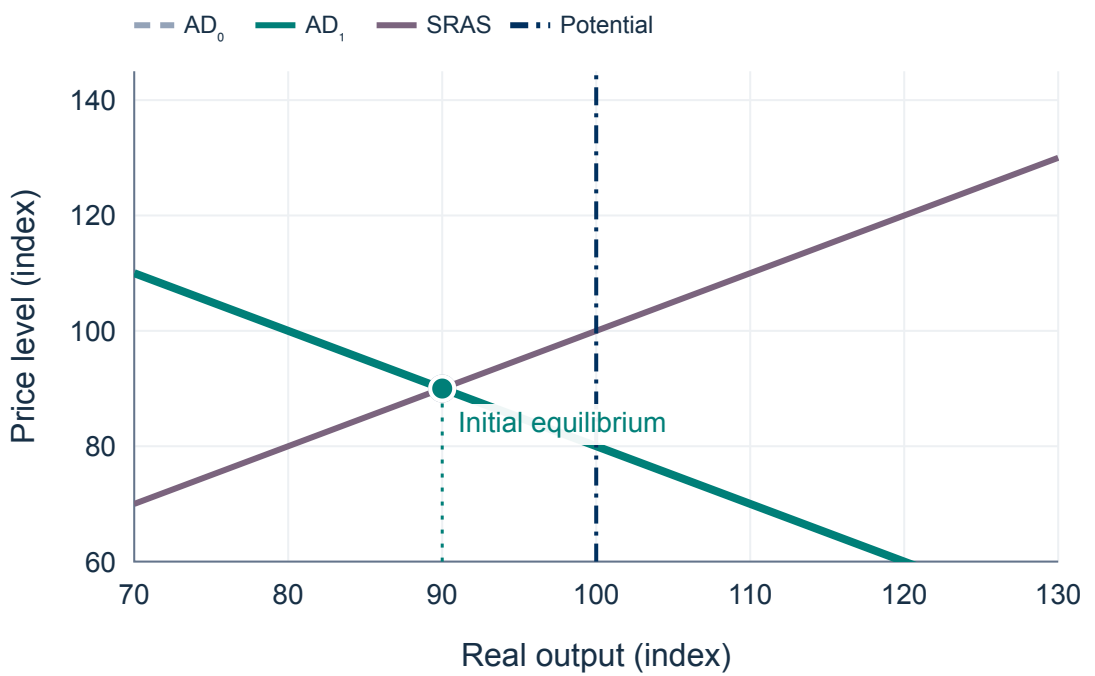
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NX: Net exports	Higher domestic spending can increase imports. Imports reduce NX, other things equal.

Expansionary Fiscal Policy

Starting output = 90 · Potential output = 100

Expansionary · AD increases



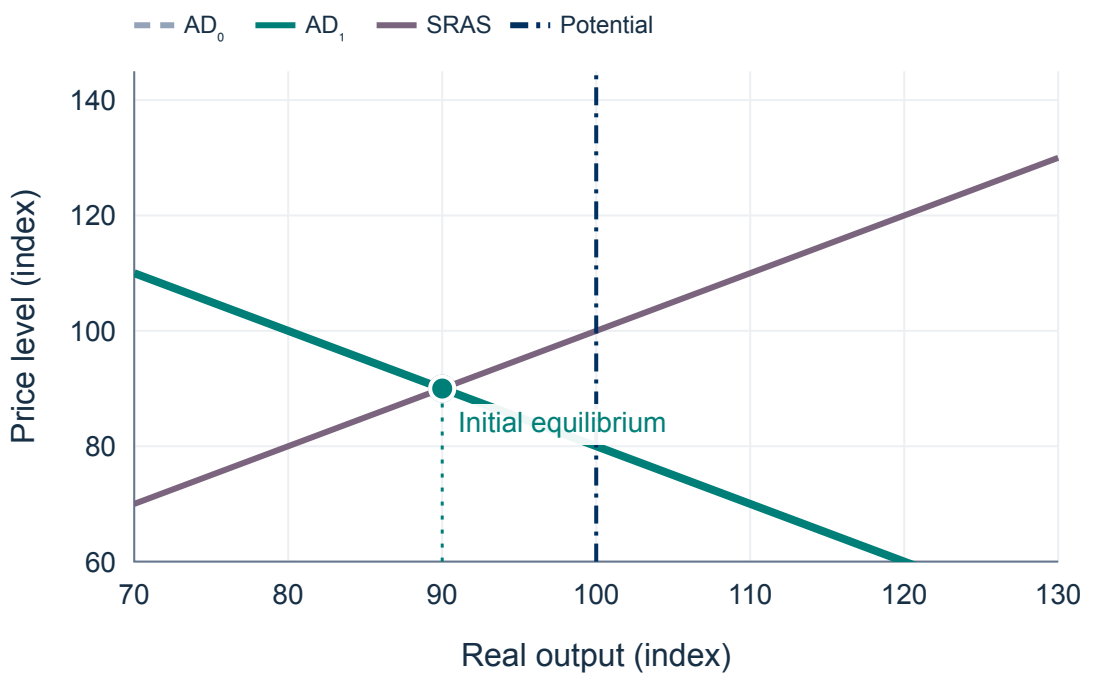
Recessionary gap

Output is below potential.

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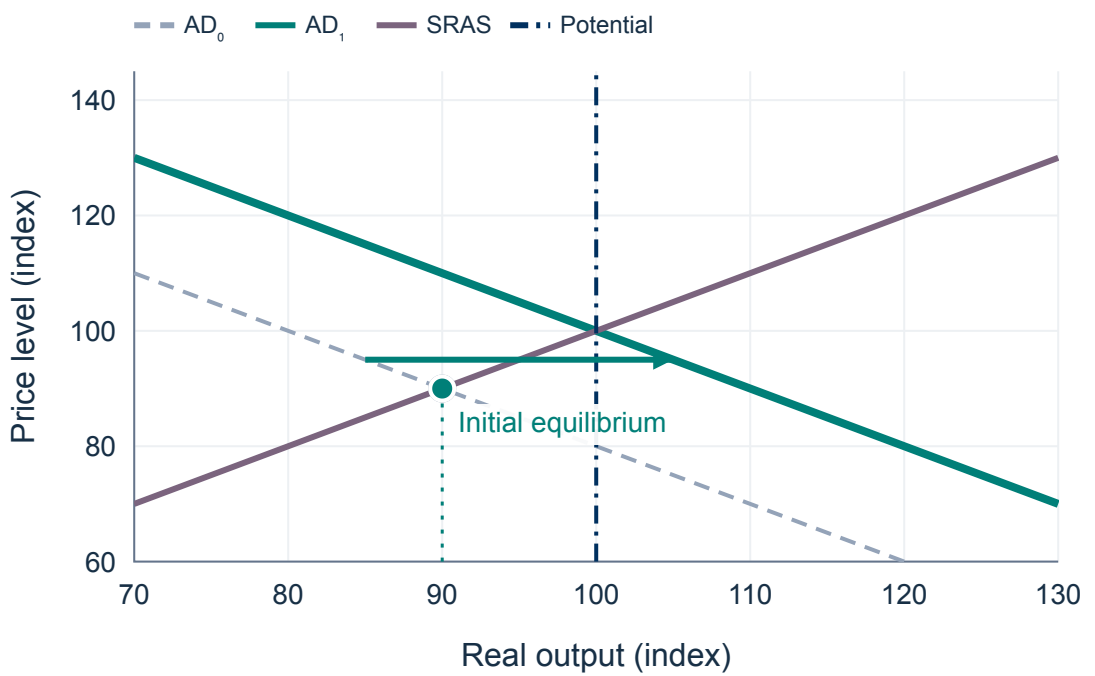
Policy change

Government purchases increase.

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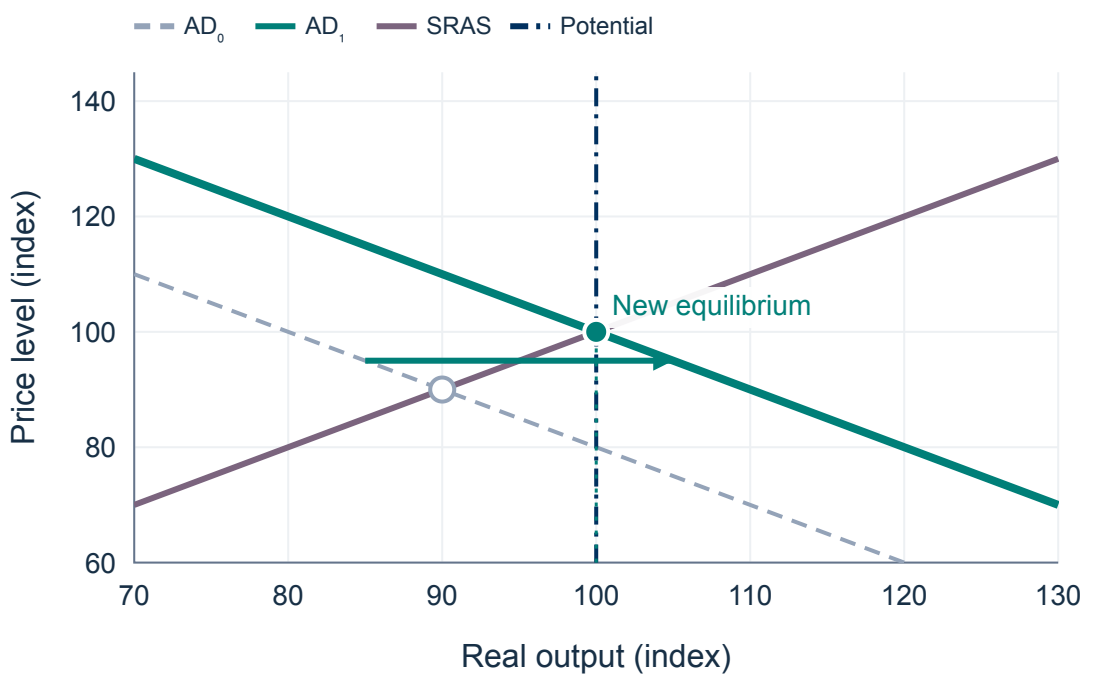
Demand response

AD shifts right; production rises.

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Demand response

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New equilibrium

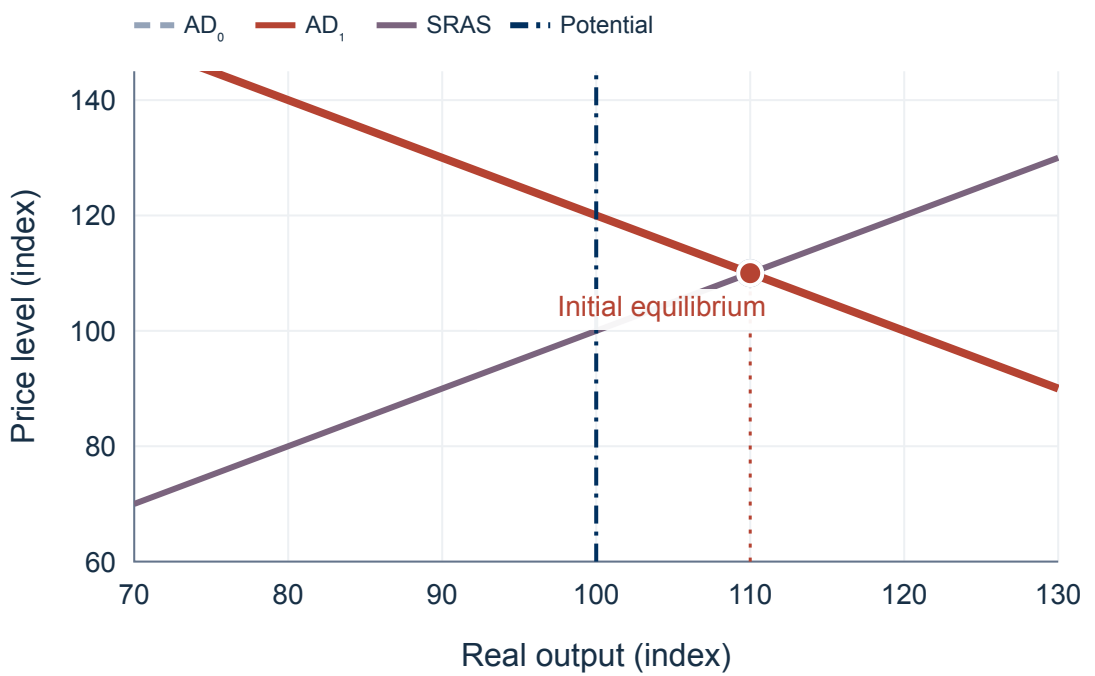
Output reaches 100; the price level rises.

Expansionary alternatives: G ↑, household T ↓, or TR ↑.

Contractionary Fiscal Policy

Starting output = 110 · Potential output = 100

Contractionary · AD decreases



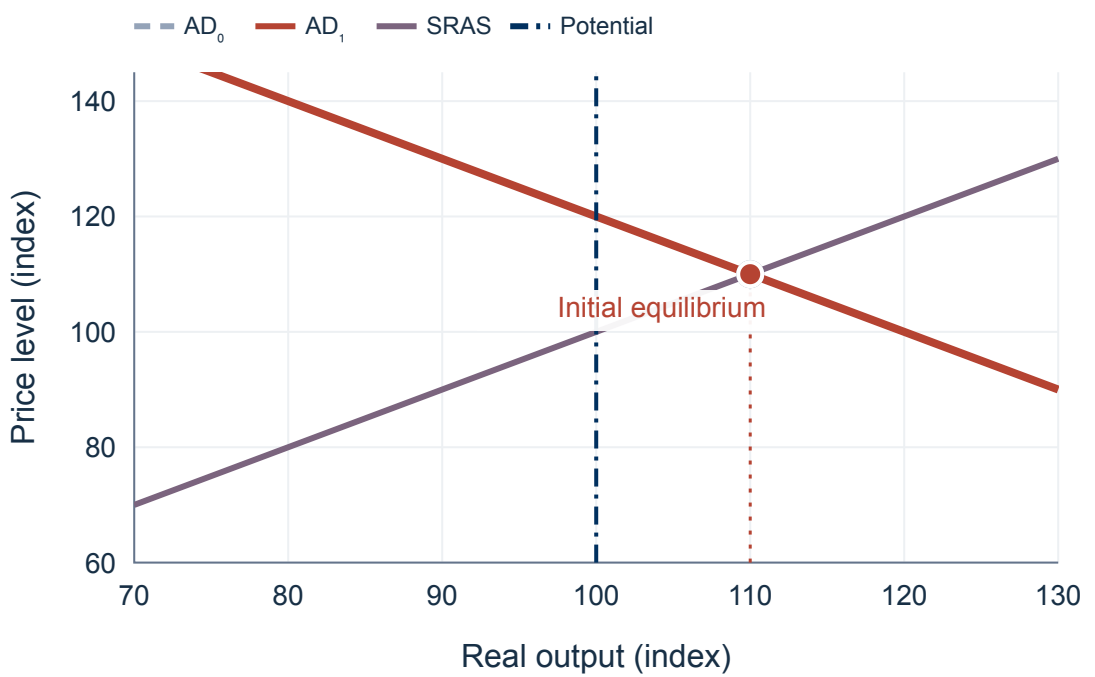
Inflationary gap

Output is above potential.

Contractionary Fiscal Policy

Starting output = 110 · Potential output = 100

Contractionary · AD decreases



Inflationary gap

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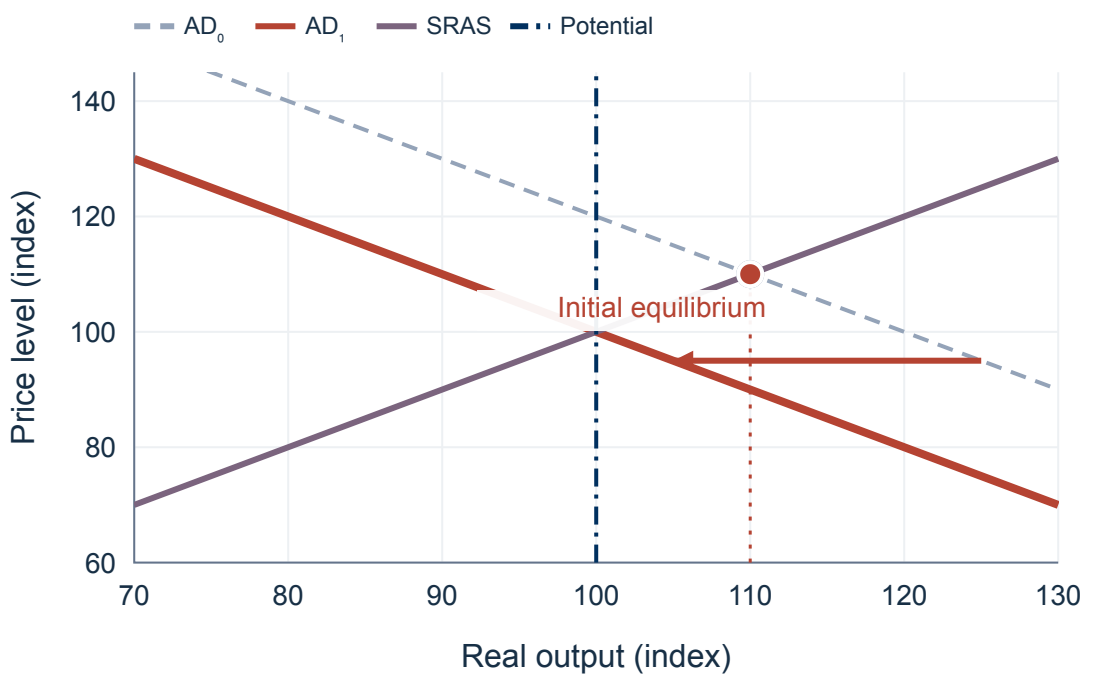
Policy change

Government purchases decrease.

Contractionary Fiscal Policy

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Contractionary · AD decreases



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Policy change

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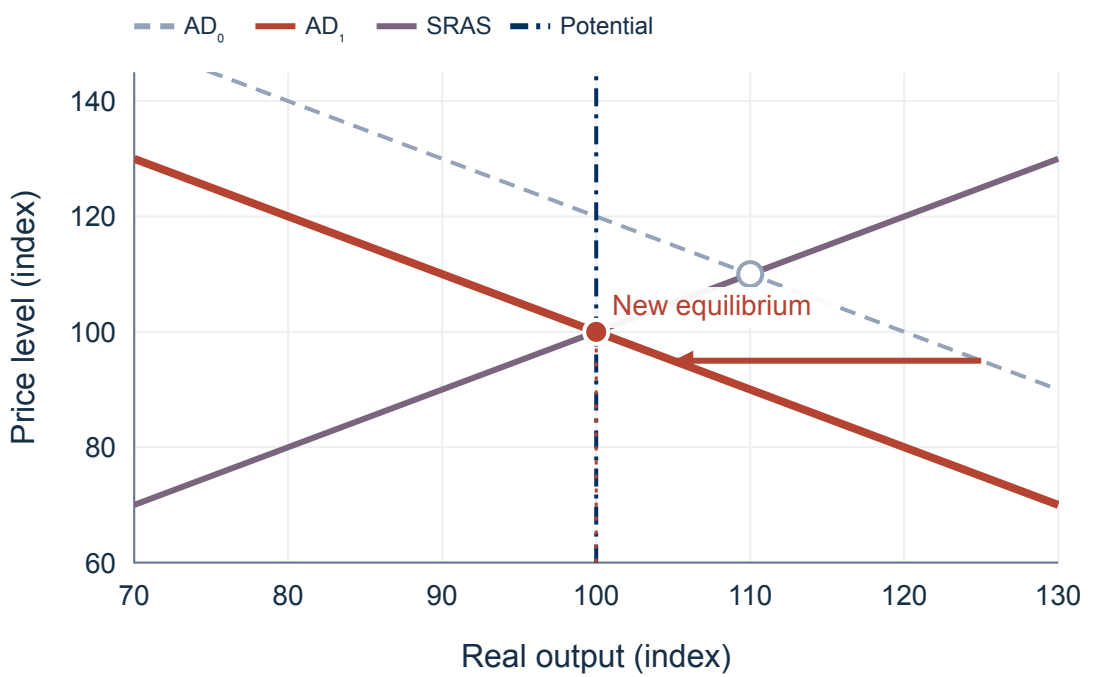
Demand response

AD shifts left; production falls.

Contractionary Fiscal Policy

Starting output = 110 · Potential output = 100

Contractionary · AD decreases



Inflationary gap

Output is above potential.

Policy change

Government purchases decrease.

Demand response

AD shifts left; production falls.

New equilibrium

Output returns to 100; the price level falls.

Contractionary alternatives: G ↓, household T ↑, or TR ↓.

Fiscal Policy Checkpoint

Identify the policy direction and the immediate spending channel.

A. Output is below potential; government repairs more roads.

B. Output is above potential; household income taxes rise.

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Expansionary. Purchases directly increase G.

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Contractionary. Disposable income and consumption fall.

Three Criticisms of Fiscal Expansion

Criticism	Mechanism	Qualification
Investment crowding out	Borrowing can raise interest rates and reduce I.	The size depends on economic and monetary conditions.
Consumption offset		
Ricardian equivalence		

An offset can weaken a policy without eliminating its effect.

References: CBO fiscal-policy framework; CRS fiscal-policy review

Three Criticisms of Fiscal Expansion

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Consumption offset	Current tax financing can reduce disposable income and C.	The reduction need not cancel the extra G.
Ricardian equivalence	A tax cut may be saved for expected future taxes.	A conditional tax-timing argument, holding spending fixed.

An offset can weaken a policy without eliminating its effect.

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Fiscal Policy Lags: The 2008–2009 Response

Recognition

Diagnose the shock.

15 Sep 2008

Lehman bankruptcy

26 Jan 2009

Recovery bill introduced

17 Feb 2009

ARRA signed

May 2009

$250 benefit payments

Authorization and delivery occur at different times.

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The Spending-Round Multiplier

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With MPC = 0.8, the spending-round factor is $m = 1/(1 - MPC) = 5$.

Compare the total multiplier effect

Net output change: ΔY = initial expenditure change × m.

The same m can give different output effects when initial spending differs.

Government Purchases: Successive Spending Rounds

ΔG = +$100 billion · MPC = 0.8 · Fixed price level

Government purchases

+$100 billion

Initial expenditure: ΔAE = +$100 billion.

First spending

$100 billion

Fixed taxes and transfers; no imports or investment offsets.

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Next consumption round

0.8 × $100 = $80 billion

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Further consumption rounds

$64 billion + $51.2 billion + …

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Further consumption rounds

$64 billion + $51.2 billion + …

Net output effect: ΔY = $100bn × 5 = +$500bn

Spending-round factor: m = 1/(1 − 0.8) = 5

Fixed taxes and transfers; no imports or investment offsets.

A Household Tax Cut

Household taxes fall by $100 billion · MPC = 0.8

> **Disposable income rises**
>
> $\Delta T = -\$100$ billion; $\Delta Y_D = +\$100$ billion.

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With MPC = 0.8, households consume $80 billion and save $20 billion.

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Apply the same spending-round factor m = 5

Net output effect: $\Delta Y = \$80 \text{ billion} \times 5 = +\$400$ billion.

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Keep the denominator clear

m = 400/80 = 5; output per $1 of tax cut = 400/100 = 4.

Purchases, Tax Cuts, and Transfers

MPC = 0.8 · Change one instrument at a time

Expansionary change	First channel	Initial expenditure
G increases by $100 billion	Government purchases	+$100 billion in G
Household T falls by $100 billion		
TR increases by $100 billion		

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TR increases by $100 billion	Disposable income +$100 billion	+$80 billion in C

Same spending-round factor m = 5; the policy starts with $100 in G or $80 in C.

Comparing Multiplier Effects

MPC = 0.8 · m = 1/(1 − MPC) = 5 · Fixed lump-sum taxes

Policy change ($bn)	Initial ΔAE ($bn)	Common factor m	Net ΔY ($bn)	Net ΔS_g ($bn)
ΔG = +100	+100 in G	5	+500	−100
ΔT = −100				
ΔTR = +100				

One policy change at a time; fixed prices, no imports or interest/investment offsets.

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Output multiplier effect: ΔY = initial ΔAE × m

Same m = 5. Output per $1 of policy: purchases 5; tax cuts and transfers 4.

Net budget change: $\Delta S_g = \Delta T - \Delta G - \Delta TR$.

A $500bn output gap: ΔG = $500bn ÷ 5 = $100bn.

One policy change at a time; fixed prices, no imports or interest/investment offsets.

Income Taxes and the Spending Multiplier

MPC = 0.8 · Income-tax rate t = 25% · Fixed price level

Additional output

→ →

$\Delta Y = \$100$

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MPC = 0.8 · Income-tax rate t = 25% · Fixed price level

Additional output

$\Delta Y = \$100$

$\rightarrow$

Taxes and disposable income

$\Delta T = \$25 \quad \Delta Y_D = \$75$

$\rightarrow$

Income Taxes and the Spending Multiplier

MPC = 0.8 · Income-tax rate t = 25% · Fixed price level

Additional output		Taxes and disposable income		Additional consumption
$\Delta Y = \$100$	$\rightarrow$	$\Delta T = \$25 \quad \Delta Y_D = \$75$	$\rightarrow$	$\Delta C = 0.8 \times \$75 = \$60$

Only \$60 reaches the next spending round, compared with \$80 under fixed taxes.

MPC remains 0.8; MPS remains 0.2.

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Additional output		Taxes and disposable income		Additional consumption
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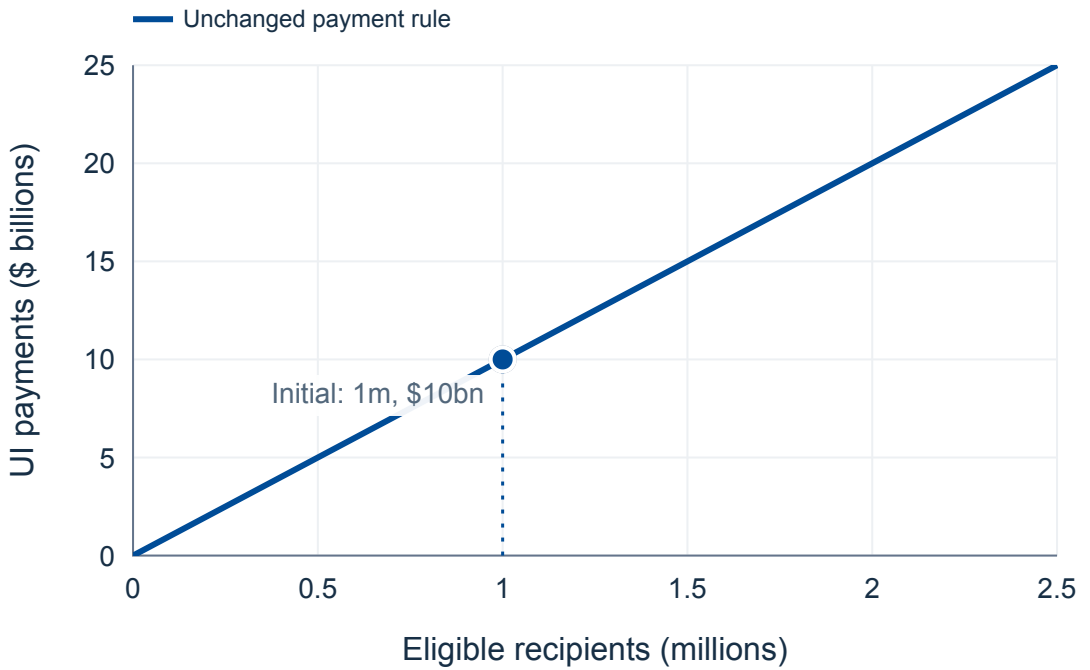
MPC remains 0.8; MPS remains 0.2.

$m = 1 / [1 - MPC(1 - t)]$

$m = 1 / [1 - 0.8(0.75)] =$ **2.5**, compared with **5** under fixed taxes.

Automatic Stabilizers

Unemployment insurance (UI): $10,000 per eligible recipient in this period



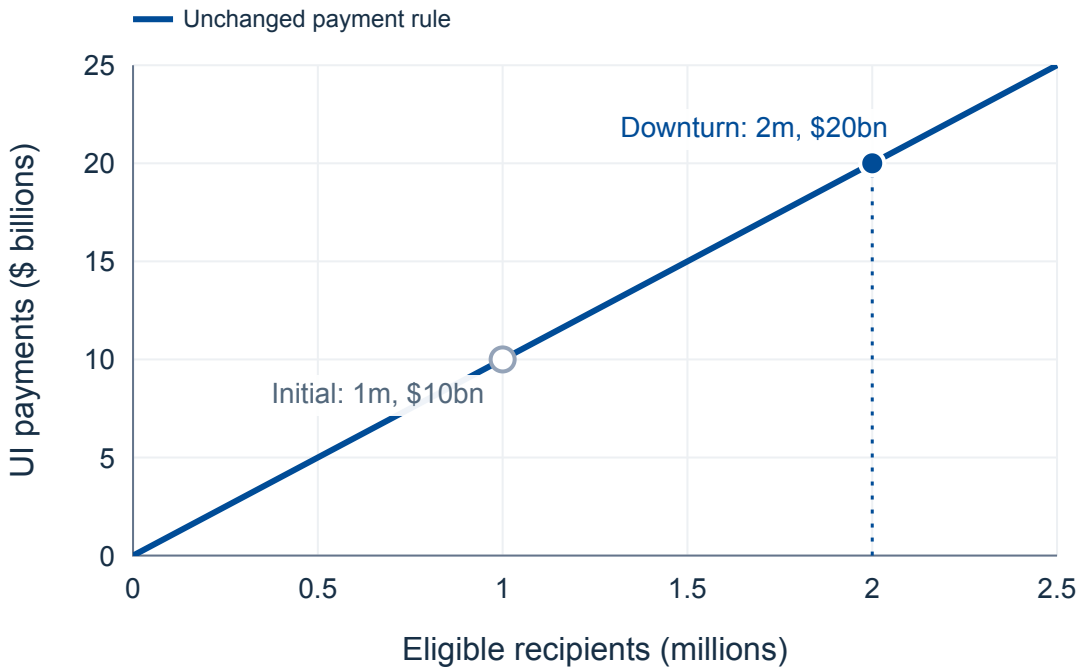
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More workers qualify under the same benefit rules.

Mechanism: CBO, Automatic Stabilizers (2024); graph values are model inputs.

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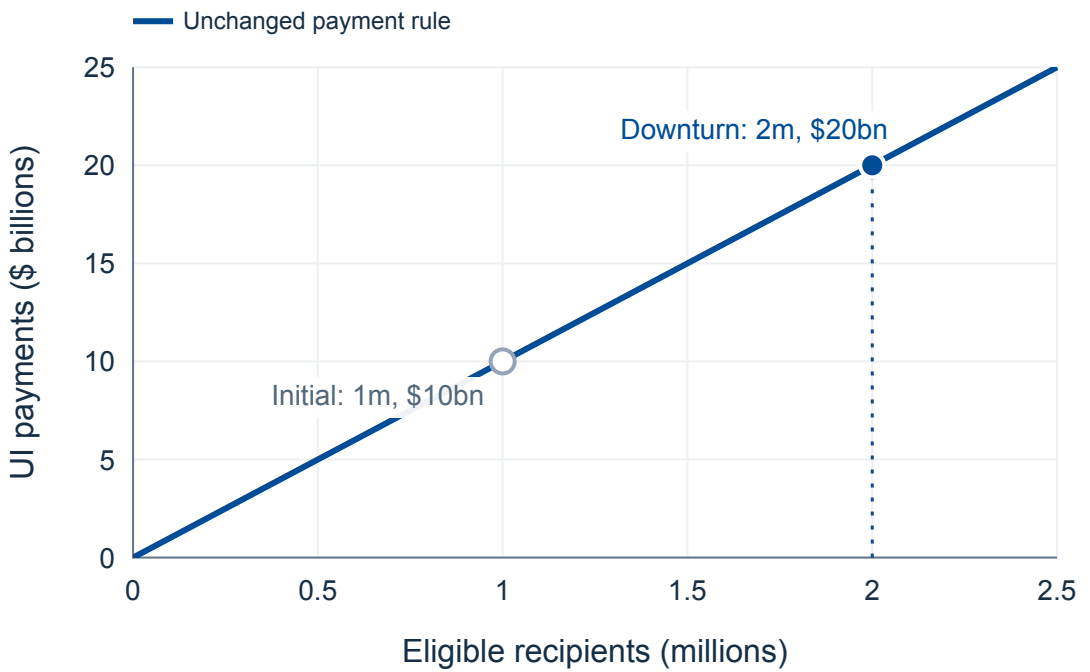
Transfers rise automatically

1m → 2m recipients; UI payments rise $10bn → $20bn.

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Consumption is cushioned

Benefits support disposable income and C; no new law is needed.

Income taxes also stabilize: falling income reduces tax payments under existing rules.

Mechanism: CBO, Automatic Stabilizers (2024); graph values are model inputs.

Government Budget Balance

$$S_g = T - G - TR$$

T: tax revenue G: purchases TR: transfers

Positive balance	Zero balance	Negative balance
Surplus	Balanced budget	Deficit

Simplified budget: debt interest and other budget items omitted.

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T = \$500bn · G = \$400bn · TR = \$150bn

S_g = \$500bn − \$400bn − \$150bn = **−\$50bn**

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$T = \$500\text{bn} \cdot G = \$400\text{bn} \cdot TR = \$150\text{bn}$

$S_g = \$500\text{bn} - \$400\text{bn} - \$150\text{bn} = \mathbf{-\$50\text{bn}}$

The balance alone does not reveal the effect on output.

Simplified budget: debt interest and other budget items omitted.

Balanced-Budget Expansion

All amounts in $ billions · ΔG = +100 · ΔT = +100 · MPC = 0.8 · Transfers unchanged

Policy change	Initial ΔAE	Policy multiplier	Total ΔY
Purchases +100	+100 in G	$k_G = 1/(1 - 0.8) = 5$	5 × 100 = **+500**
Taxes +100			
Net effect			

Fixed lump-sum taxes and prices; spare capacity; no imports or interest/investment offsets.

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Net effect	100 − 80 = **+20**	Both policy effects combined	500 − 400 = **+100**

First round: **+$20bn**. All spending rounds: ΔY = 20 × 5 = **+$100bn**.

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Budget balance: $\Delta S_g = \Delta T - \Delta G - \Delta TR = 100 - 100 - 0 =$ **$0**

Balanced but expansionary: extra taxes cover extra purchases, while real GDP rises.

Fixed lump-sum taxes and prices; spare capacity; no imports or interest/investment offsets.

The Budget and the Business Cycle

Existing tax schedules and benefit rules · No new legislation

Budget item	Activity weakens	Activity strengthens
Tax revenue	Usually falls	Usually rises
Unemployment benefits		
Budget balance		

Concept reference: CBO, Effects of Automatic Stabilizers on the Federal Budget (2024)

The Budget and the Business Cycle

Existing tax schedules and benefit rules · No new legislation

Budget item	Activity weakens	Activity strengthens
Tax revenue	Usually falls	Usually rises
Unemployment benefits	More eligible claims	Fewer eligible claims
Budget balance		

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Budget balance	Moves toward deficit	Moves toward surplus

Moving toward surplus can mean a smaller deficit.

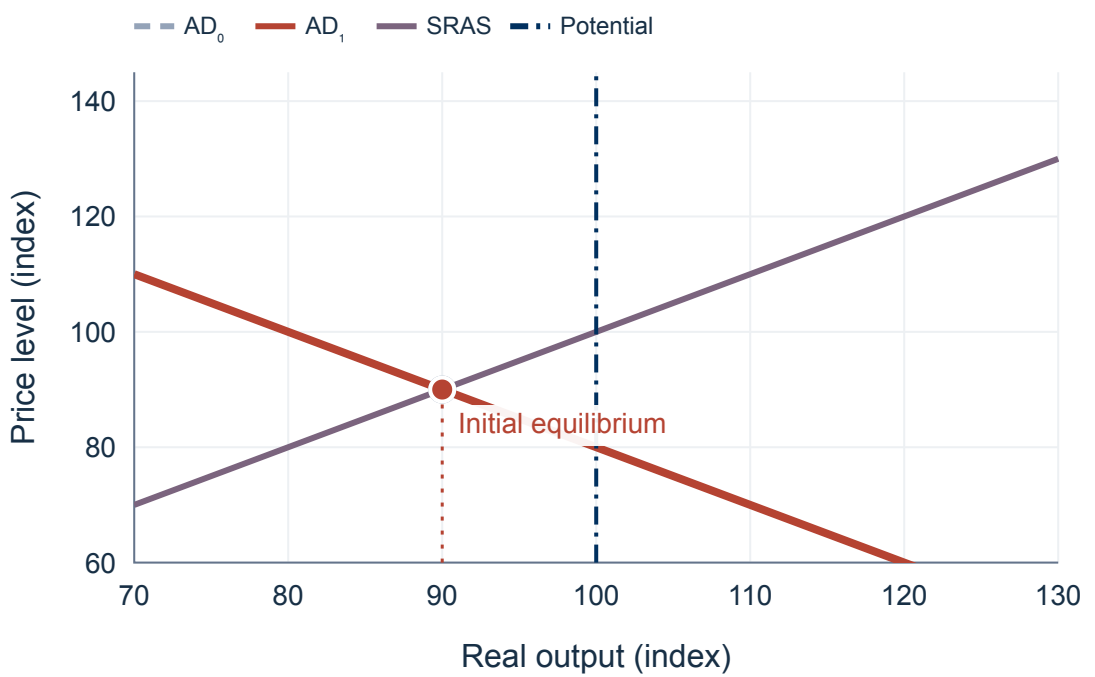
- Cyclically adjusted budget balance: estimates the balance at potential output, helping separate business-cycle effects from other changes in the budget.

Concept reference: CBO, Effects of Automatic Stabilizers on the Federal Budget (2024)

Annual Budget Balance and Stabilization

Demand recession · Starting output = 90 · Potential output = 100

Contractionary · AD decreases



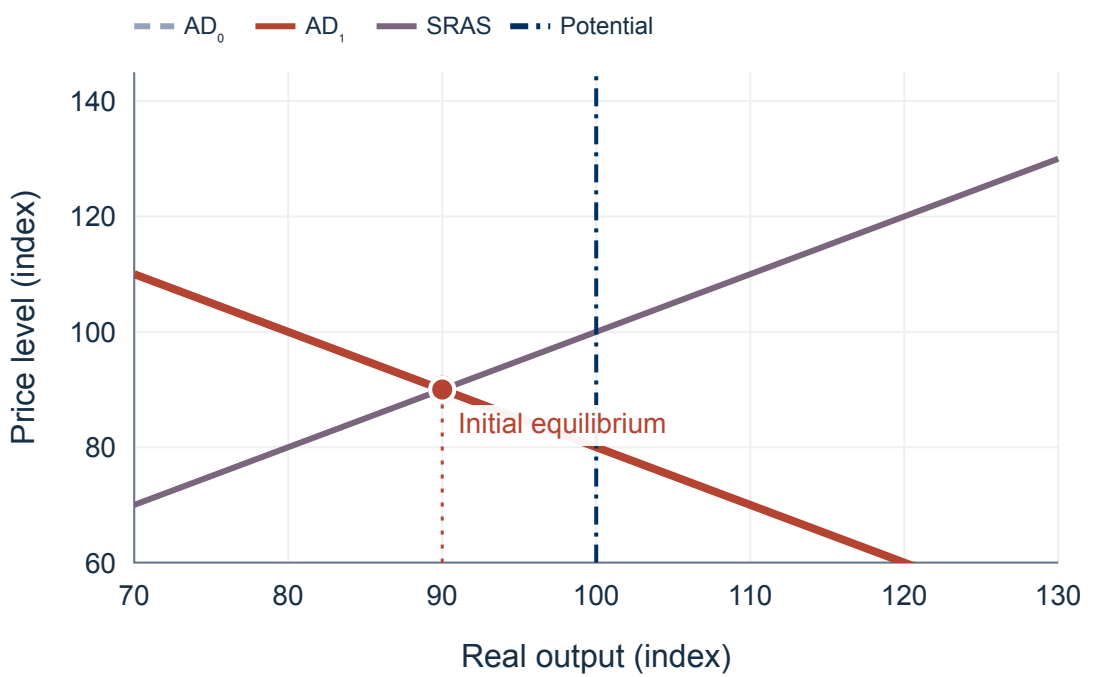
Initial recession

Output is already below potential.

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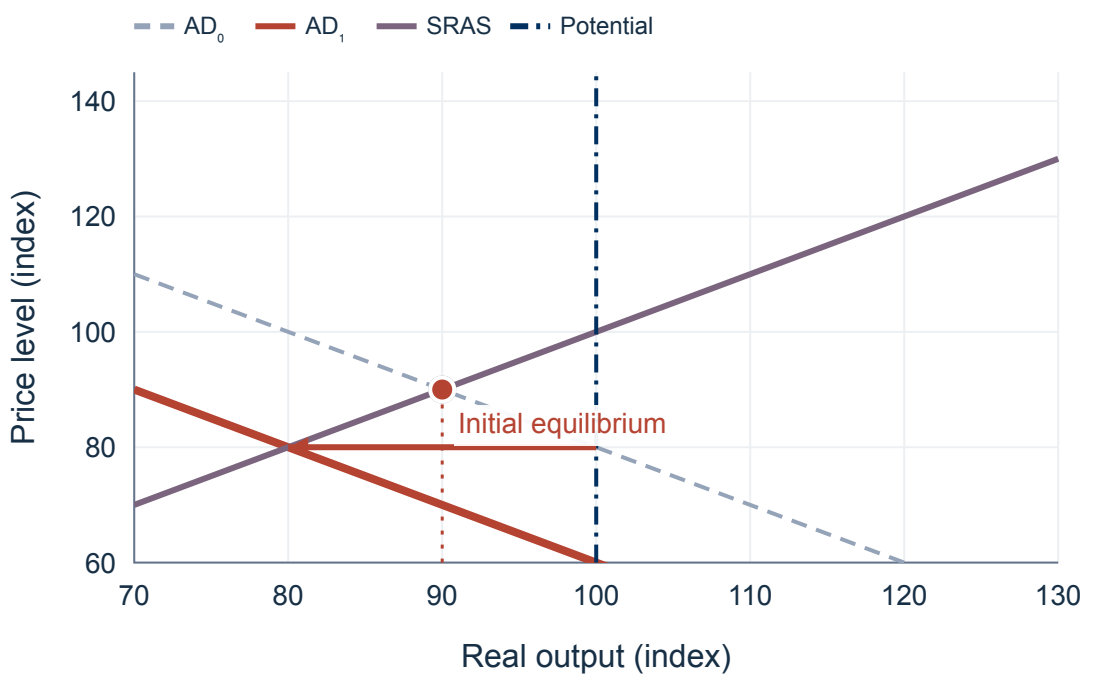
Annual-balance requirement

Falling receipts trigger tax increases or spending cuts.

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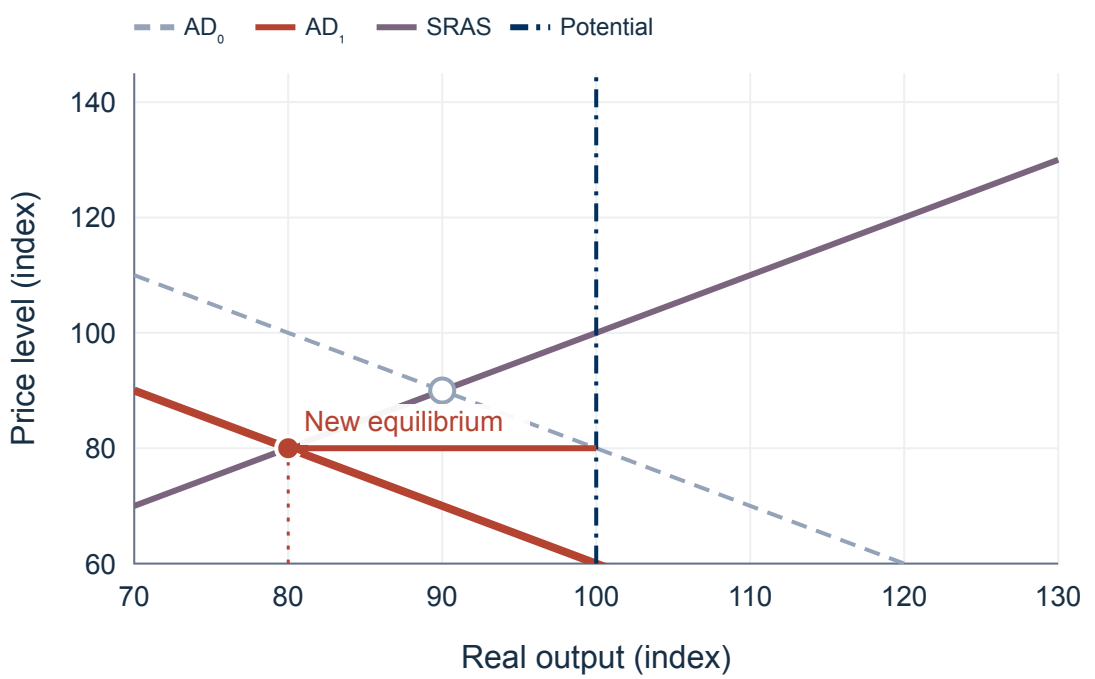
Demand response

The offset reduces AD further.

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Annual-balance requirement

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Demand response

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Deeper output gap

Output falls from 90 to 80.

A strict annual-balance rule can work against automatic stabilizers.

Fiscal Stabilization and Sustainability

Downturns

Fiscal space allows stabilizers and temporary support to operate.

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Stronger economic activity

An opportunity to reduce deficits and rebuild fiscal space.

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An opportunity to reduce deficits and rebuild fiscal space.

Debt over time

Assess debt relative to GDP, interest costs, growth, and future balances.

A deficit is an annual flow; debt is a stock accumulated over time.

Fiscal Policy: Final Checkpoint

Explain the mechanism and the evidence you would need.

A. Income falls; the tax schedule stays unchanged.

B. A new law raises weekly unemployment benefits.

C. With MPC = 0.8, compare a $100 purchase and a $100 tax cut.

D. The deficit rises in a recession. Does this prove new stimulus?

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Check new laws, automatic stabilizers, and the cyclically adjusted balance.

Fiscal Policy Cheat Sheet: Demand

Expenditure: C + I + G + NX. Purchases enter G; household taxes and transfers influence C; business taxes can influence I.

Policy	Main mechanism	Curve	Short-run result
Increase purchases	Government buys more current goods and services.	AD right	Output and price level rise.
Cut household taxes	Disposable income rises; households spend part.	AD right	Output and price level rise.
Increase transfers	Recipients spend part of the extra income.	AD right	Output and price level rise.
Cut purchases or transfers; raise household taxes	Spending falls.	AD left	Output and price level fall.

First three: expansionary. Fourth: contractionary. Transfers themselves are not G.

Accounting: BEA · Demand mechanism: CBO

Fiscal Policy Cheat Sheet: Supply

Fiscal policy can also affect firms' costs and the economy's productive capacity.

Policy	Main mechanism	Possible curve shifts	Conditional result
Productive infrastructure, education or research	Purchases support demand now; successful investment improves capacity later.	AD right initially; possibly SRAS and LRAS right later.	Potential output can rise.
Lower production taxes or subsidize inputs	Lower marginal production costs can encourage supply.	Potentially SRAS right.	Output rises and price level falls from the supply effect alone.

Design and financing matter. Public spending must improve productivity to raise capacity.

Fiscal framework: CBO, How CBO Analyzes Fiscal Policies

Demand Shocks: Adjustment and Policy

Y* = potential output. Automatic adjustment below assumes no policy response and unchanged potential.

Initial shock	Immediate result	Automatic adjustment	Fiscal response to the gap	Policy consequence
AD left	Y < Y*; price level falls.	Wages and costs fall; SRAS moves right.	Expansionary policy moves AD right.	Supports output and offsets some downward price pressure.
AD right	Y > Y*; price level rises.	Wages and costs rise; SRAS moves left.	Contractionary policy moves AD left.	Returns output toward potential and eases price pressure.

Self-adjustment changes SRAS through wages and costs; fiscal stabilization usually changes AD.

Supply Shocks: Adjustment and Policy

Y* = potential output. These two cost shocks leave potential unchanged; adjustment assumes no policy response.

Initial shock	Immediate result	Automatic adjustment	Fiscal response to the gap	Policy consequence
SRAS left	Y < Y*; price level rises.	Lower wages or reversal of the cost shock can move SRAS right.	Expansionary policy moves AD right.	Supports output, but raises the price level further.
SRAS right	Y > Y*; price level falls.	Higher wages or reversal of the favorable shock can move SRAS left.	Contractionary policy moves AD left.	Returns output toward potential, but lowers the price level further.

After an adverse supply shock, demand policy faces an output–price tradeoff.

Potential Output and Policy Tradeoffs

Productive capacity changes

LRAS shifts; SRAS may also shift. Adjustment is toward the new potential output.

Choose the relevant benchmark

Fiscal choices depend on the remaining gap and the policy objective. Restoring the old GDP level may be inappropriate.

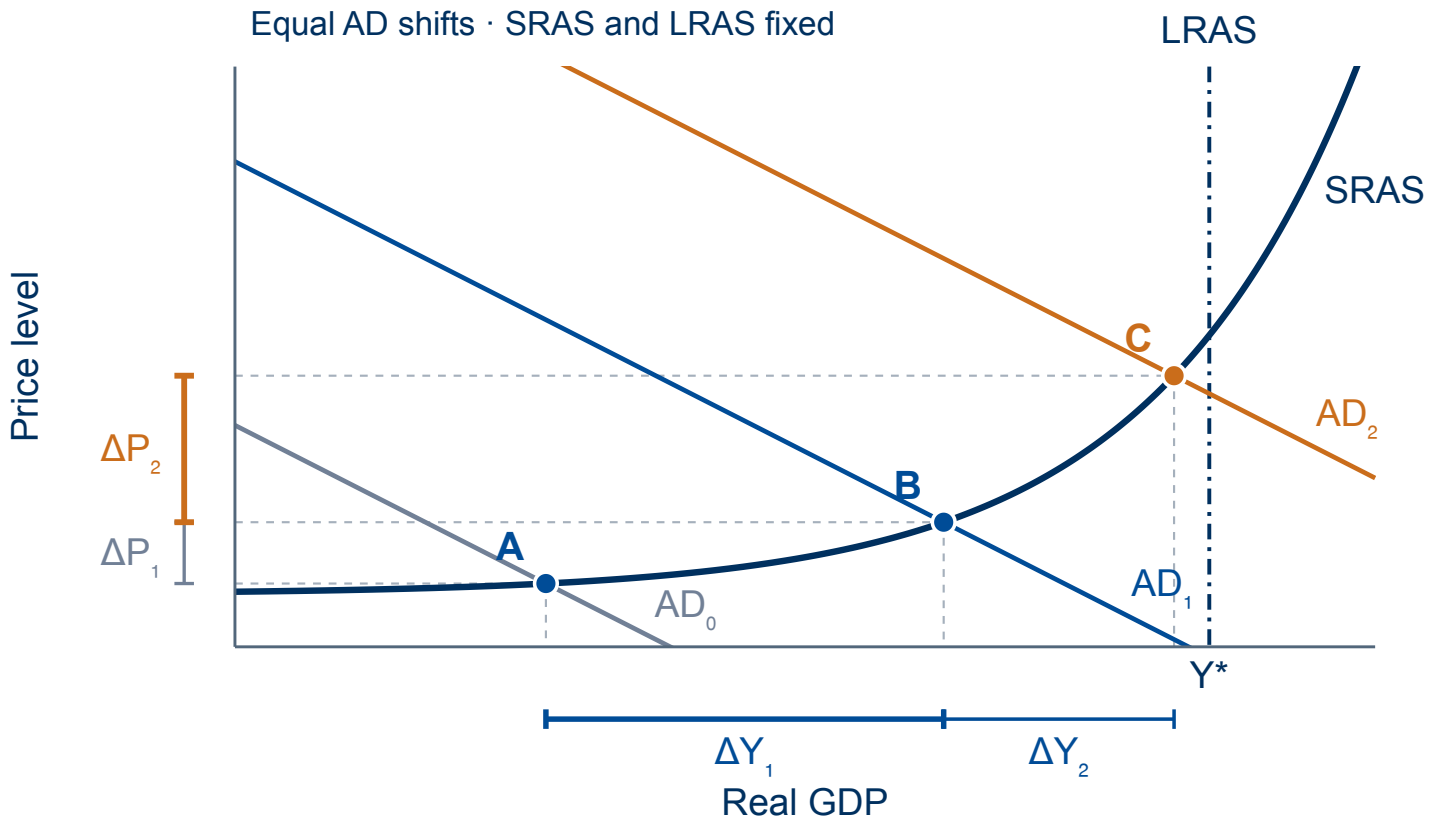
Keep the two mechanisms distinct

Wage and cost adjustment shifts SRAS; fiscal stabilization usually shifts AD.

Results depend on timing, financing, household spending, productivity, and central-bank responses.

The limits of continued fiscal expansion

As spare capacity disappears, additional demand generates smaller output gains and stronger inflation pressure.



Substantial spare capacity

Firms employ idle resources. Output rises considerably, with limited price pressure.

Capacity becomes tighter

Overtime, scarce workers, and production bottlenecks raise costs. Output responds less and prices respond more.

Continued expansion

Persistent demand growth beyond productive capacity strengthens inflation pressure. Higher interest rates may further reduce the output response.

Fiscal expansion becomes less effective at increasing real output as supply constraints tighten.

Oxford College of Emory University

Chapter 13, Part 2

The Long-Run Implications of Fiscal Policy

ECON_OX 112 · Principles of Macroeconomics

Guilherme Paiva Pinto

Roadmap

Budget balances and financing

Deficits, public debt, and refinancing risk.

History and sustainability

Greece, U.S. data, and debt relative to GDP.

Investment and commitments

Productive capacity, interest, and future spending.

Money and banks

Real resource limits, followed by Chapter 14.

Deficits, Surpluses, and Public Debt

> **Deficit: a flow**
>
> Spending, including net interest, exceeds revenue during a period.

Definitions: U.S. Treasury, public debt FAQs

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> **Surplus: a flow**
>
> Revenue exceeds spending during a period.

> **Debt: a stock**
>
> Outstanding borrowing at a date; debt held by the public excludes federal accounts' holdings.

In a simplified budget, next year's debt = this year's debt + the total deficit.

Definitions: U.S. Treasury, public debt FAQs

Debt Service and Refinancing Risk

> **Debt service**
>
> Interest payments use revenue or require additional borrowing.

Framework: CBO, economic effects of fiscal policy

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Maturing debt must be repaid or replaced; lenders may demand higher rates or stop lending.

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Possible feedback

Higher rates can increase borrowing and later interest costs; this is not automatic.

Crowding out affects private investment; debt service and refinancing affect the public budget.

Framework: CBO, economic effects of fiscal policy

Greece: The Sovereign Debt Crisis

> **2009: fiscal credibility**
>
> Revisions revealed earlier misreporting and much larger fiscal imbalances.

Sources: IMF, Greece report 10/110; European Commission, assistance chronology

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Borrowing became harder and more expensive as investor confidence fell.

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May 2010: assistance and adjustment

Euro-area partners and the IMF supported a program tied to fiscal adjustment and reforms.

Greece used the euro; its national government did not control an independent currency.

Sources: IMF, Greece report 10/110; European Commission, assistance chronology

The Debt-to-GDP Ratio

All amounts in $ billions · Debt and GDP use current dollars

Measure	Initial year	Following year
Debt at year-end	600	620
Annual nominal GDP	1,000	1,050
Debt / GDP × 100		

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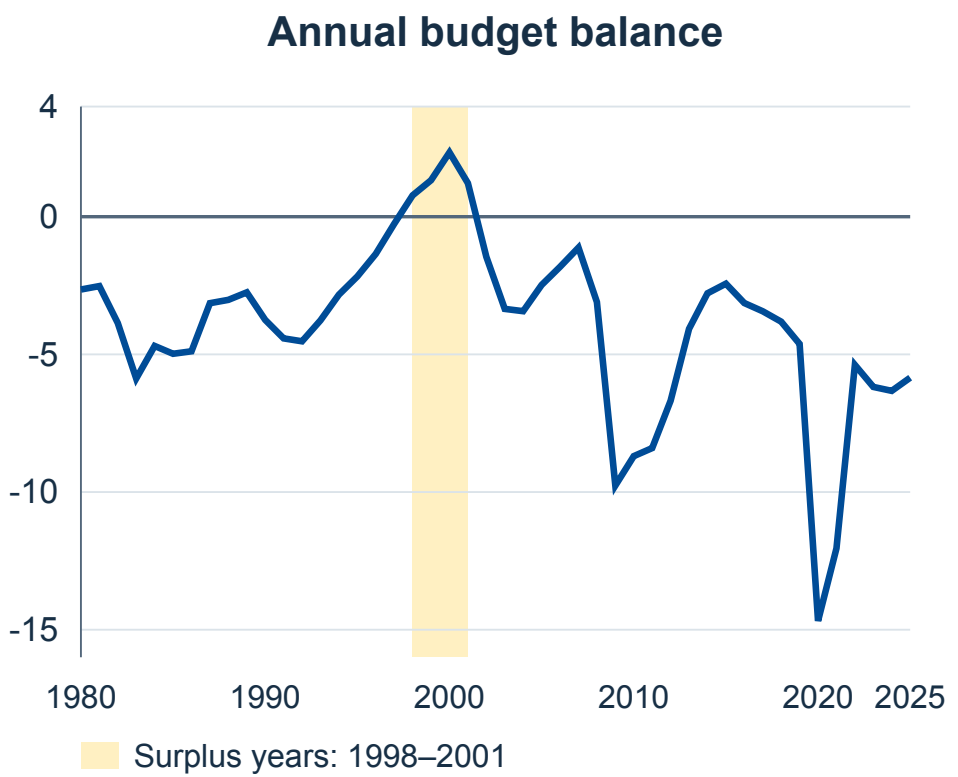
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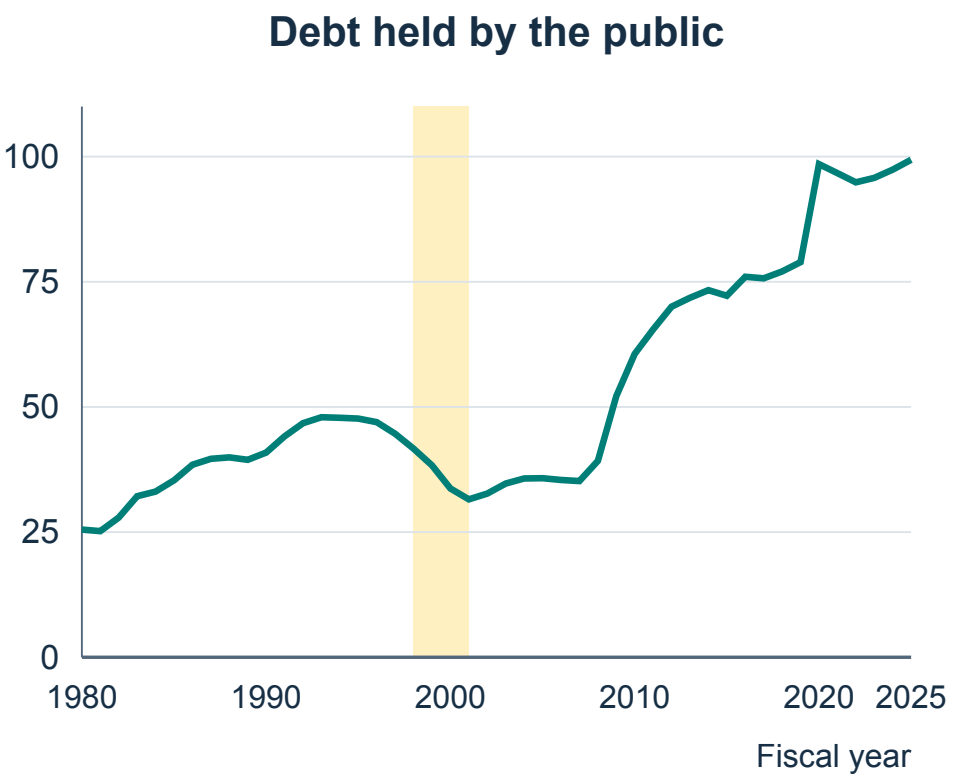
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Debt rises by 20, but GDP grows faster in percentage terms, so the ratio falls.

U.S. Budget Balances and Public Debt

Federal government · Fiscal years 1980–2025 · Percent of GDP





Data: CBO, Historical Budget Data, February 2026; original course chart

Borrowing, Investment, and Growth

Private investment

With other saving unchanged, government borrowing can raise financing costs and reduce private investment.

Framework: CBO, economic effects of fiscal policy

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Future productive capacity

Less capital accumulation can lower future productivity and potential output.

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Future productive capacity

Less capital accumulation can lower future productivity and potential output.

Public investment

Effective infrastructure, education, or research can support capacity; design and financing determine the net effect.

A short-run increase in demand and a long-run increase in productive capacity are different claims.

Framework: CBO, economic effects of fiscal policy

Interest Costs and the Primary Balance

Amounts in $ billions · Beginning debt = 600 · Annual interest rate = 4% · Primary deficit = 10

> **Primary deficit**
>
> Noninterest spending minus revenue. The total deficit also includes net interest.

Budget concepts: CBO, Budget and Economic Outlook, February 2026

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0.04 × 600 = 24.

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Interest cost

0.04 × 600 = 24.

Total deficit

24 + 10 = 34.

End-of-year debt = 600 + 34 = 634. More interest narrows other budget choices.

Budget concepts: CBO, Budget and Economic Outlook, February 2026

Implicit Liabilities and Future Commitments

Expected future program spending is distinct from outstanding Treasury debt.

Social Security

Retirement, disability, and survivor benefits; population ageing affects financing.

Fiscal outlook: CBO, Budget and Economic Outlook, February 2026

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Health coverage mainly for older people; ageing and medical costs affect spending.

Medicaid

Health coverage linked to eligibility, enrollment, costs, and policy choices.

Mandatory spending follows existing law; future costs depend on policy and assumptions.

Fiscal outlook: CBO, Budget and Economic Outlook, February 2026

Long-Run Fiscal Policy Checkpoint

Explain the distinction before giving a policy conclusion.

A. The annual deficit falls from 40 to 20 and remains positive. What happens to debt?

B. A deficit finances a new transport network. Does higher G prove a permanent increase in potential output?

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No. Capacity gains depend on usefulness, implementation, maintenance, and financing costs.

Fiscal Financing and Money Creation

A different financing method

Issuing money changes financial claims; it does not create workers, machinery, or goods.

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Real resource limits

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Institutional limits

Greece shared the euro; Treasury borrowing and central-bank operations are different actions.

Chapter 14: What is money, who creates it, and what role do banks play?